universität wien
Faculty of Psychology

Lernen unter COVID-19-Bedingungen

Parents/Legal guardians

Method Report

1. Introduction

The present cross-sectional study is the second phase of the project "Learning under COVID-19 conditions" with pupils and students previously archived at https://doi.org/10.11587/VRQL3B and at https://doi.org/10.11587/XIU3TX, respectively.

The second phase of the project comprises three additional target groups – parents, teachers and school principals, in one cross-sectional measurement point. Thus, this phase of the project aims to investigate behaviour and coping of other agents in the school system, beside pupils', in order to get a better picture of the situation and effects of COVID-19 on the education system in Austria.

Study Description

1.1. Title and version

Lernen unter COVID-19-Bedingungen: Eltern (Learning under COVID-19 conditions: parents, V1.0)

1.2. Principal Investigators

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1.3. Funding/Acknowledgments

The present study has been funded by the Vienna Science and Technology Fund (WWTF), by MEGA Bildungsstiftung, and the City of Vienna through project COV20-025.

1.4. Fieldwork agency

The fieldwork of the study "Lernen unter COVID-19-Bedingungen" was managed and handled internally.

1.5. Topics/Keywords

- Sociodemographic information: age, sex, education, language, region
- Situation at home: employment, income, work-mode during pandemic
- Parental role: parental role overload, household task division, support, self-efficacy
- Effects of crisis: health-, financial-, parenting-related stress
- Contact with school: institutional trust, communication patterns,

- Relationship with child/ren: open communication, perception of overload in kids, trustworthiness in the child & positive developments

2. Study Design

2.1. Sampling

The sample of the study "Lernen unter COVID-19-Bedingungen" consisted of parents or legal guardians whose child/ren attended secondary education in Austria. The parents were invited to participate in the survey by headmasters of the schools that their child/ren attended. Moreover, the link to and a recommendation to participate in the survey was published by the Federal Ministry of Education, Science, and Research on their website. The link with a respective invitation to the study was also placed on the survey's website. In order to participate in the study, the respondents had to agree to the informed consent, be Austrian residents, and have child/ren who attend(s) secondary school. It is important to note that the sample cannot be considered representative for the Austrian population as many students do not have easy access to the internet and/or devices needed to fill out the survey. Therefore, risk groups may be underrepresented in the current sample.

2.2. Fieldwork

The overall data collection was conducted in the time from 06 April 2021 to 25 May 2021.

3. Data Cleaning

3.1. General data cleaning procedure

The cleaning procedure contained two steps:

3.1.1. Initial data cleaning

As participating parents answered an online questionnaire via Unipark (Questback, 2007), data was exported from unipark in .csv format and imported to SPSS. Variables names were replaced. All missing values were defined as -99. Invalid zero values or values of -77 were replaced with -99. Participants who gave consent and had answering time (variable **duration**) larger than zero were selected for a new SPSS file.

3.1.2. Data cleaning

As unipark assigns -1 to variable duration in cases where the break was made during the answering the questionnaire, all valid -1 cases were replaced with (+) 9999999 value. Participants with 100% missings on main outcome variables (related to self-efficacy) were excluded. Minimum possible duration was tested by filling in the questionnaire as fast as possible while reading all items. All cases above the duration treshold and above 18 years og age were selected for a new SPSS file.

3.1.3. Missing values

All empty spaces were replaced with -99. Missing values that were represented by 0 in the raw data set are replaced by -99 on all variables where 0 is not a valid value. Missing values represented with -77 in unipark were replaced with -99 as well. Missing values on variables that were result of filters applied, were defines with -98. Cases where participants by mistake got to see a question they should not have seen (due to technical issues in unipark) were replaces with -78.

3.1.4. Filters

Data records of participants who did not give their consent are deleted.

A fastest possible time for answering the questionnaire is determined diligently. Variable duration of answers with duration below the fastest possible time is set to 9999999.

Data records with 100% missings on main outcome variables are deleted.

All other filters are defined in the questionnaire, as well as in data set using values -98 and -78 (see above) as value labels.

3.1.5. Answers to open questions

All open questions have been removed from the data sets due to anonymization rules.

3.1.6. Multiple choice / single choice

Question "Von wem bekommen Sie bei diesen Aufgaben Unterstützung?" (Partner/in, Eltern/Schwiegereltern, andere Familienmitglieder, Freund/innen, Nachbar/innen, Kinder) was posed in the multiple choice format.

3.2. Anonymisation procedure

3.2.1. Code variable

Due to high risk for identification, variables ep.code (personal code), ep.child.code1 (identifier for matching the children of the same parents), ep.child.code2 (identifier for matching the children of the same parents), ep.child.code3 (identifier for matching the children of the same parents), ep.child.code4 (identifier for matching the children of the same parents), ep.child.code5 (identifier for matching the children of the same parents) were deleted from the data set.

3.2.2. Federal state variables

In this dataset, for "ep.child.fs", answers "Burgenland" and "Niederösterreich" are merged to "Burgenland/Niederösterreich", answers "Tirol" and "Vorarlberg" are merged to "Tirol/Vorarlberg", and answers "Kärnten" and "Steiermark" are merged to "Kärnten/Steiermark".

3.2.3. School type variables

In the school type variable (ep.child.school), school types that occur rarely are categorized to "Andere" to get groups with at least 20 observations.

For ep.child.bhsbms.grade ("In welche Klasse geht Ihr Kind? (Wenn "Berufsbildende höhere Schule (BHS)" oder "Berufsbildende mittlere Schule (BMS) ")"), answers "4. Klasse / 12. Schulstufe" and "5. Klasse / 13. Schulstufe" were merged to get groups with more than 20 observations.

As there were fewer than 20 answers for ep.child.bsle.grade ("Wie weit ist Ihr Kind in der Lehre? (Wenn "Berufsschule und Lehre")"), it was removed from the dataset.

3.2.4. Language variables

In this dataset, the selected display languages other than German (ep.language) are categorized to "andere" to get groups with at least 20 observations.

3.2.5. Gender variables

In this dataset, answers indicating that the corresponding person chose "divers" and "möchte ich nicht sagen" (ep.gender) are recoded to missing as there are fewer than 20 observations.

3.2.6. Age variables

In this dataset, participants with the age of 34 years old or younger as well as 56 years old or older were merged together to get groups with at least 20 observations for ep.age.

In this dataset, answers regarding the age of the participants' children were merged to get age cohorts with at least 20 observations for ep.child.age1 ("Alter des ersten Kindes"), ep.child.age2 ("Alter des zweiten Kindes"), ep.child.age3 ("Alter des dritten Kindes") and ep.child.age4 ("Alter des vierten Kindes"). Moreover, as there were fewer than 20 answers for ep.child.age5 ("Alter des fünften Kindes"), ep.child.age6 ("Alter des sechsten Kindes"), ep.child.age7 ("Alter des siebten Kindes"), and no answers for ep.child.age8 ("Alter des achten Kindes"), ep.child.age9 ("Alter des neunten Kindes"), ep.child.age10 ("Alter des zehnten Kindes") these variables were removed from the dataset.

3.2.7. Closed questions

The following answer categories were merged in this dataset to get groups with at least 20 observations:

Variable(s)	Answers merged	New value label
ep.family.employ	all numerical values indicating 5 or more	5 oder mehr
ep.ger.frequency	• selten • nie	selten oder nie
ep.homesup3	• überwiegend jemand anderer • jemand anderer als ich	überwiegend jemand anderer oder jemand anderer als ich
ep.childnr	all numerical values indicating 4 to 10	4 bis 10

Additionally, the following answer categories were merged:

Variable(s)	Answers merged	New value label
ep.ger.support.appointments	• fast immer • immer	fast immer oder immer
ep.ger.docs	• eher schlecht • schlecht	eher schlecht oder schlecht
ep.ger.teacher	• eher schlecht • schlecht	eher schlecht oder schlecht
ep.ger.news	• eher schlecht • schlecht	eher schlecht oder schlecht

3.2.8. Open questions

In this dataset, five open variables are deleted without recoding or standardizing the answers:

- ep.child.relationship.other ("In welchem Verhältnis stehen Sie zu dem Kind, für das Sie den Fragebogen ausfüllen? Andere:")
- ep.child.school.other ("Welche Schule besucht das Kind, für das Sie diesen Fragebogen ausfüllen? Andere")
- ep.childcare.why.other ("Haben Sie während der letzten Distance Learning Phase (d.h. als der Unterricht ausschließlich online stattfand) Ihr Kind zur Betreuung in die Schule gegeben? Warum? Andere")
- ep.challenges ("Was waren seit Beginn der Pandemie die größten Herausforderungen für Sie?")
- ep.accomplish ("Was ist Ihnen seit Beginn der Pandemie besonders gut gelungen?")

3.2.9. Open questions containing numeric values

Answers to questions requiring numeric values are recoded according to the description in section "Data cleaning procedure". Original (string) answers are deleted.

3.3. Further deletions

3.3.1. Consent variable

One variable containing the participants' agreement on processing the filled in data for scientific purposes is deleted (ep.consent) as all included participants gave their consent for processing the filled in data for scientific purposes.

3.3.2. Filter variables

One filter variable (filter_$) originating from the cleaning procedure is deleted.

3.3.3. Variables with few valid values

In this dataset, one variable that contains only missing values (quality) is deleted.

3.3.4. System variables

System variables that contain information of the duration of answering separate pages are deleted.

- external_lfdn (external sequence number)
- tester (filter variable indicating entries made for test purposes)
- dispcode (disposition code)
- lastpage (identifier of exit page)
- ep.consent
- Language (language)
- browser (browser identifier)
- referer (HTTP referer)
- page_history (array of page identifiers indicating the sequence of visited questionnaire pages)
- hflip (questionnaire answered on a screen in landscape mode)
- vflip (questionnaire answered on a screen in portrait mode)
- session_id (session identifier)
- ats (absolute timestamp)
- datetime (date of first access)
- date_of_last_access (date of last access)
- date_of_first_mail (date of first e-mail)
- rts4959071, rts4960151, rts4960152, rts4960156, rts4960157, rts4960161, rts4960162, rts4960163, rts4960164, rts4960165, rts4960166, rts4960168, rts4964310, rts4964324, rts4964325, rts4964409, rts4964425, rts4964427, rts4964451, rts4964546, rts4964551, rts4964553, rts4964560, rts4964564, rts4964576, rts4965375, rts4965391, rts4965401, rts4965404, rts4965427, rts4965438, rts4965457, rts4965747, rts4983396, rts4983397, rts4983770, rts4988751 (relative timestamps for pages)

3.3.5. Reverse-coded variables

All reverse-coded variables are deleted.

- ep.income.change.rec (Reversed Likert scale of item ep.income.change)
- ep.ger.frequency.rec (Reversed Likert scale of item ep.ger.frequency)
- ep.ger.docs.rec (Reversed Likert scale of item ep.ger.docs)

- ep.ger.teacher.rec (Reversed Likert scale of item ep.ger.teacher)
- ep.ger.news.rec (Reversed Likert scale of item ep.ger.news)
- ep.ger.support.measures.rec (Reversed Likert scale of item ep.ger.support.measures)
- ep.ger.support.appointments.rec (Reversed Likert scale of item ep.ger.support.appointments)
- ep.lockdown.rec (Reversed Likert scale of item ep.lockdown)
- ep.family.lockdown.rec (Reversed Likert scale of item ep.family.lockdown)
- ep.child.lockdown.rec (Reversed Likert scale of item ep.child.lockdown)
- ep.schooltrust1.rec (Reversed Likert scale of item ep.schooltrust1)
- ep.fc4.rec (Reversed Likert scale of item ep.fc4)

3.3.6. Mean scale variables

- ep.hrs (Mean scale Health-related stress)
- ep.fs (Mean scale Financial stress)
- ep.roleover (Mean scale Roleoverload)
- ep.socsup (Mean scale Social support)
- ep.homesup (Mean scale Help with household tasks)
- ep.communication (Mean Kommunikation/Koordination mit anderen Eltern, Schule)
- ep.parse (Mean Parenting self-efficacy)
- ep.ag (Mean scale Agency)
- ep.co (Mean scale Communion)
- ep.optimism (Mean scale Optimism)
- ep.child.reliab (Mean scale Zuverlässigkeit des Kindes & Positive Entwicklungen)
- ep.child.comm (Mean scale Open communication)
- ep.child.strain (Mean scale Einschätzung Belastung des Kindes)
- ep.schooltrust (Mean scale Vertrauen in Schule (insgesamt))
- ep.schooltrust.trust (Mean scale Vertrauen in Schule (nur Vertrauen items 1-3))
- ep.schooltrust.info (Mean scale Vertrauen in Schule (nur Info items 4-5))
- ep.fc (Mean scale Concerns about kids future)

3.4. Increasing reusability

To increase the reusability of the data set variables are rearranged into an appropriate order. Variable names are renamed to get more meaningful names.